

MERIDIAN

Latent Cartographer

Local semantic embeddings inside a Chromium extension

FOUNDATION MILESTONE

Manifest V3 scaffold, packaged MiniLM model, dedicated WASM worker and offline verification

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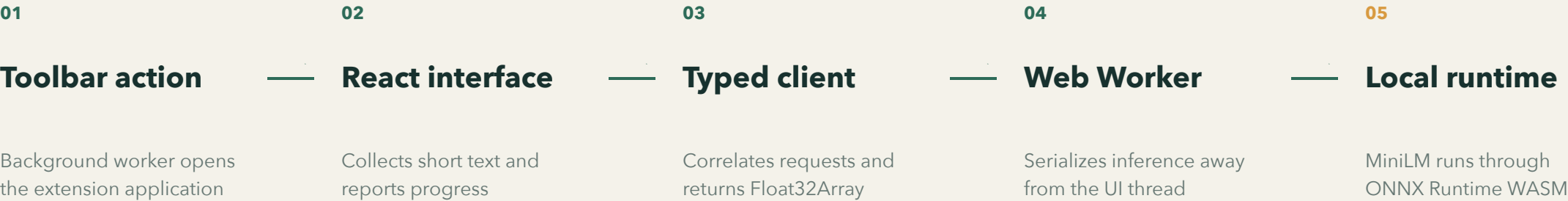
M1

384

embedding dimensions

LOCAL MODEL
NO API KEY
WASM CPU

The implemented embedding path



Runtime boundary

The model, tokenizer and WASM files load from the installed extension. The embedding request does not require a backend, account or API key.

chrome-extension://

single extension origin

Reproducible build and offline controls

BUILD PATH

01 — Install pinned dependencies

npm ci restores the versions recorded in package-lock.json.

02 — Prepare local assets

The script downloads one pinned MiniLM revision and validates assets.lock.json.

03 — Build the extension

Vite emits the UI, background worker, embedding worker, model and WASM into dist/.

04 — Load unpacked in Chromium

Manifest V3 starts the diagnostic interface from the packaged output.

OFFLINE CONTROLS



Remote models disabled

Transformers.js can only resolve packaged model files.



Origin guard

Fetch requests outside the extension origin fail immediately.



Strict CSP

connect-src 'self' blocks external runtime connections.



Compatible WASM mode

One thread runs without cross-origin isolation. Proxy mode stays off.

A real 384-dimensional embedding generated locally

VERIFIED OUTPUT

384

dimensions

WASM

CPU backend

7.6 ms

warm inference

1.000000

normalized vector

MERIDIAN / MILESTONE 1

A local foundation for your research.

Generate a real semantic embedding with the model packaged inside this extension.

Embedding diagnostic

MiniLM · 384 dimensions · WASM CPU · External model loading disabled

Text to embed

Database transactions provide atomicity and isolation.

Short text only in this milestone. Long inputs may be truncated by the model; document chunking comes later.

Generate embedding

Embedding generated locally.

Dimensions	Backend	Elapsed	Vector norm
384	wasm	7.6 ms	1.000000

► Inspect all vector values

Offline execution and semantic separation

TEST BOUNDARY

- 01 Fresh Playwright Chromium profile
- 02 Offline mode enabled before the app opens
- 03 Unreachable proxy with DNS resolution disabled
- 04 HTTP, HTTPS and WebSocket requests rejected

The check runs inside the browser. It does not claim firewall isolation across the entire machine.

MEASURED RUN

360.2 ms

first call with model initialization

7.7 / 7.6 ms

later calls reuse the worker model

Related text



cosine 0.588948

Unrelated text



cosine -0.069150

The model placed related sentences closer than an unrelated sentence in the verified run.

Foundation complete, product workflow still ahead

READY NOW

- 01

—

Manifest V3 extension

Toolbar action, strict CSP and Vite production build
- 02

—

Local embedding worker

Pinned MiniLM model through Transformers.js and ONNX WASM
- 03

—

Typed module boundary

EmbeddingClient and shared contracts support parallel work
- 04

—

Offline integration test

Real browser run verifies local requests and semantic output

NEXT MILESTONES

- 01

—

Capture and extraction

Web pages first, then PDF and Markdown import
- 02

—

Persistence and retrieval

IndexedDB metadata with OPFS for larger assets
- 03

—

Search and canvas

Query embeddings, cosine ranking and spatial workspace
- 04

—

Integrated workflow

Connect capture, storage, retrieval and the canvas

MODULAR TEAM SPLIT

Foundation: manifest, build, workers, contracts

Extraction: src/extraction/

Canvas UI: src/ui/